USL Assignment 8

Agglomerative Clustering algorithm

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Agglomerative Clustering Theory

1. Introduction to Agglomerative Clustering

Agglomerative Clustering is a type of hierarchical clustering algorithm used in unsupervised machine learning to group similar data points into clusters.

- It follows a bottom-up approach
- Each data point starts as an individual cluster
- Clusters are merged step by step until a single cluster is formed

Unlike K-Means, it does not rely on centroids but instead builds a hierarchical structure of clusters.

2. Key Concepts in Agglomerative Clustering

1. Linkage Criteria:

Linkage defines how the distance between clusters is calculated:

- **Single Linkage:** Distance between closest points
- **Complete Linkage:** Distance between farthest points
- **Average Linkage:** Average distance between points
- **Ward Linkage:** Minimizes variance within clusters

2. Distance Metric

- Measures similarity between data points
- Common metrics:
  - Euclidean Distance
  - Manhattan Distance

3. Dendrogram

A dendrogram is a **tree-like diagram** that shows how clusters are merged step by step.

- X-axis → Data points
- Y-axis → Distance between clusters
- Cutting the dendrogram at a certain level gives the number of clusters

3. Working of the Agglomerative Clustering Algorithm

The algorithm works as follows:

1. **Start with each data point as a separate cluster**
2. **Compute the distance between all clusters**
3. **Merge the two closest clusters**
4. **Update the distance matrix**
5. **Repeat steps 2–4 until only one cluster remains**
6. **Form final clusters by cutting the dendrogram at a chosen level**

4. Example of Agglomerative Clustering

Consider a dataset of wine samples with different chemical properties:

- Initially, each sample is its own cluster
- Similar samples are merged based on distance
- Gradually, clusters form representing different wine types
- The dendrogram visually shows this merging process

5. Advantages of Agglomerative Clustering

- Does not require predefining the number of clusters
- Produces a hierarchical structure of clusters
- Works well for small to medium-sized datasets
- Flexible due to different linkage methods

6. Limitations of Agglomerative Clustering

- Computationally expensive ($O(n^2)$)
- Not suitable for very large datasets
- Once clusters are merged, they cannot be split
- Sensitive to noise and outliers

7. Comparison with K-Means

Agglomerative Clustering and K-Means are both clustering techniques but differ in their approach and characteristics.

Agglomerative Clustering is a hierarchical method that follows a bottom-up approach, where each data point starts as an individual cluster and clusters are merged step by step. In contrast, K-Means is a centroid-based algorithm that partitions the dataset into a predefined number of clusters using iterative optimization.

Agglomerative Clustering does not require the number of clusters to be specified initially, as the number of clusters can be determined by cutting the dendrogram at a desired level. On the other hand, K-Means requires the user to specify the number of clusters (k) beforehand.

In terms of cluster shape, Agglomerative Clustering can form clusters of arbitrary shapes depending on the linkage method used, whereas K-Means generally forms spherical clusters due to its reliance on centroids.

Agglomerative Clustering produces a dendrogram that shows the hierarchical relationship between clusters, while K-Means directly outputs cluster labels without any hierarchical structure.

Overall, Agglomerative Clustering is more flexible and informative, whereas K-Means is faster and more efficient for large datasets.

8. Performance Evaluation Metrics

To evaluate clustering performance:

- **Silhouette Score** → Measures how well clusters are separated
- **Davies-Bouldin Index** → Lower value indicates better clustering
- **Adjusted Rand Index (ARI)** → Compares predicted clusters with true labels
- **Normalized Mutual Information (NMI)** → Measures shared information
- **Homogeneity** → Each cluster contains only one class
- **Completeness** → All points of a class are in one cluster
- **V-measure** → Harmonic mean of homogeneity and completeness

9. Conclusion

Agglomerative Clustering is a powerful hierarchical clustering technique that builds clusters step by step by merging similar data points. It provides a clear visualization of cluster formation through dendrograms and is useful for understanding the structure of data. However, it is computationally expensive and best suited for smaller datasets.